

Date: 30/07/2025

Lab Practical #09:

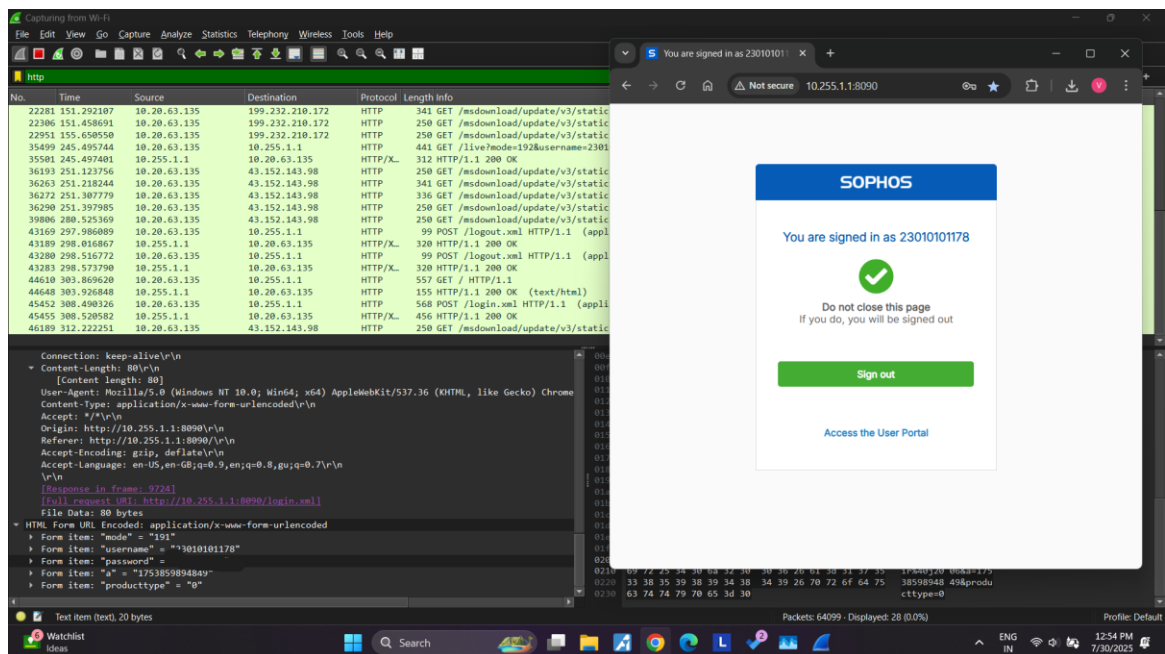
Study Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)

Practical Assignment #09:

1. Explain usage of Wireshark tool.

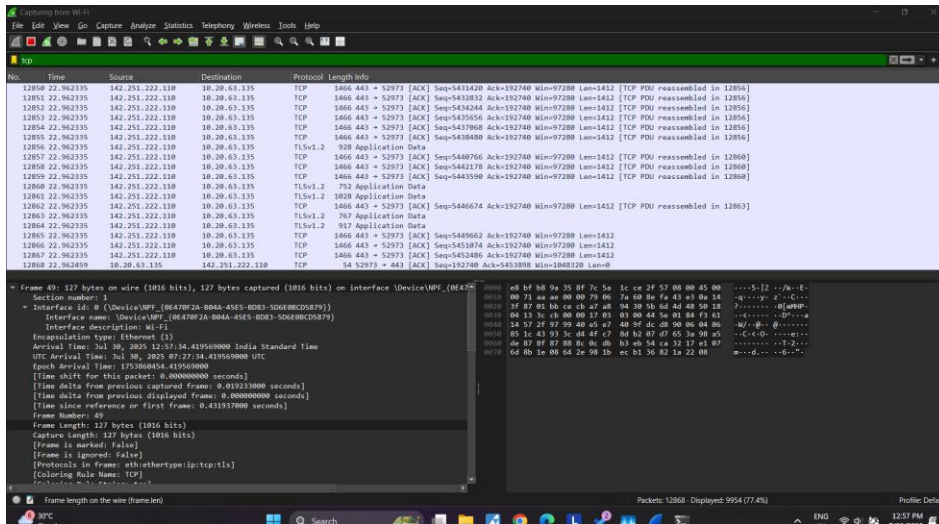
Wireshark is a tool that captures and shows the data moving in a network in real time. It is mainly used for checking network problems, keeping an eye on performance, finding possible security issues, and understanding how different network protocols work.

2. Packet capture and header analysis by Wireshark (HTTP, TCP, UDP, IP, etc.)

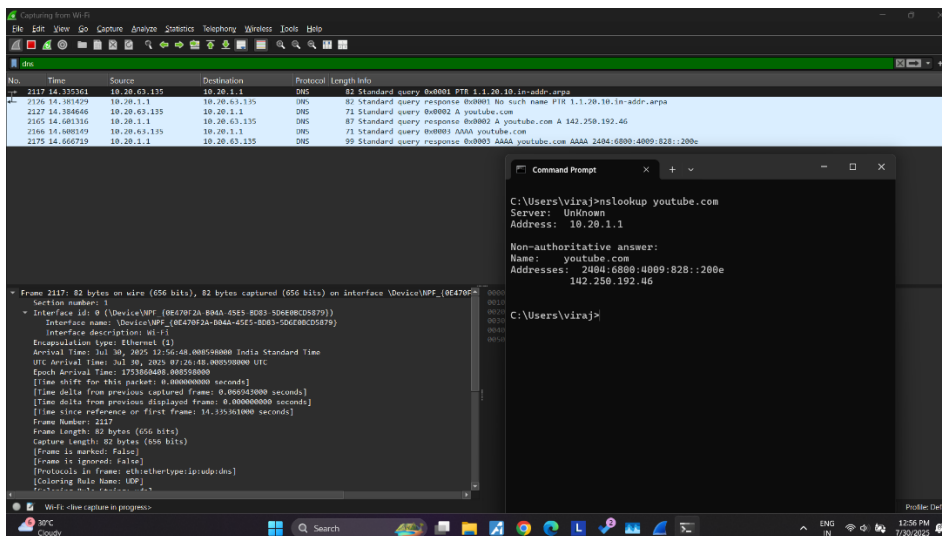


The screenshot displays the Wireshark network protocol analyzer interface. The top pane shows a list of captured packets, with the selected packet being an HTTP GET request from 10.255.1.1 to 10.255.1.1. The middle pane shows the details of the selected packet, including the request headers and the body. The bottom pane shows the raw packet data in hexadecimal and ASCII. The browser window in the background shows a Sophos login page with a 'Sign out' button.

Date: 30/07/2025



Wireshark capture showing an HTTP GET request. The packet list shows a single packet (No. 49) of type HTTP, source 10.20.63.135, destination 10.20.63.135. The packet details pane shows the structure of the HTTP packet, including the GET method and the requested resource. The packet bytes pane shows the raw data of the packet.



Wireshark capture showing a DNS query and response. The packet list shows two packets (No. 2117 and 2118) of type DNS, source 10.20.63.135, destination 10.20.1.1. The packet details pane shows the structure of the DNS packets, including the query and response. The packet bytes pane shows the raw data of the packets.

Windows Command Prompt showing the output of the nslookup command:

```
C:\Users\viraj>nslookup youtube.com
Server: Unknown
Address: 10.20.1.1

Non-authoritative answer:
Name: youtube.com
Addresses: 2484:6880:4089:828::200e
142.250.192.46
```



DARSHAN INSTITUTE OF ENGINEERING & TECHNOLOGY

Semester 5th | Practical Assignment | Computer Networks (2301CS501)

Date: 30/07/2025

The image shows a Wireshark capture of ICMP ping requests. The packet list shows several ping requests and replies. The packet details pane shows the structure of an ICMP Echo (ping) request. The packet bytes pane shows the raw data of the ping request. A Command Prompt window is open, showing the results of a ping command to youyube.com. The output shows that the ping was successful, with a round trip time of 19ms.

```
C:\Users\viraj>ping youyube.com
Ping request could not find host youyube.com. Please check the
name and try again.

C:\Users\viraj>ping youtube.com

Pinging youtube.com [142.250.192.46] with 32 bytes of data:
Reply from 142.250.192.46: bytes=32 time=19ms TTL=116
Reply from 142.250.192.46: bytes=32 time=22ms TTL=116
Reply from 142.250.192.46: bytes=32 time=22ms TTL=116
Reply from 142.250.192.46: bytes=32 time=35ms TTL=116

Ping statistics for 142.250.192.46:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 35ms, Average = 24ms

C:\Users\viraj>
```

The image shows a Wireshark capture of UDP traffic. The packet list shows several UDP packets. The packet details pane shows the structure of a UDP packet. The packet bytes pane shows the raw data of the UDP packet. The packet list shows a series of UDP packets, including a query and a response. The packet details pane shows the structure of a UDP packet, including the source and destination ports. The packet bytes pane shows the raw data of the UDP packet, including the query and response data.